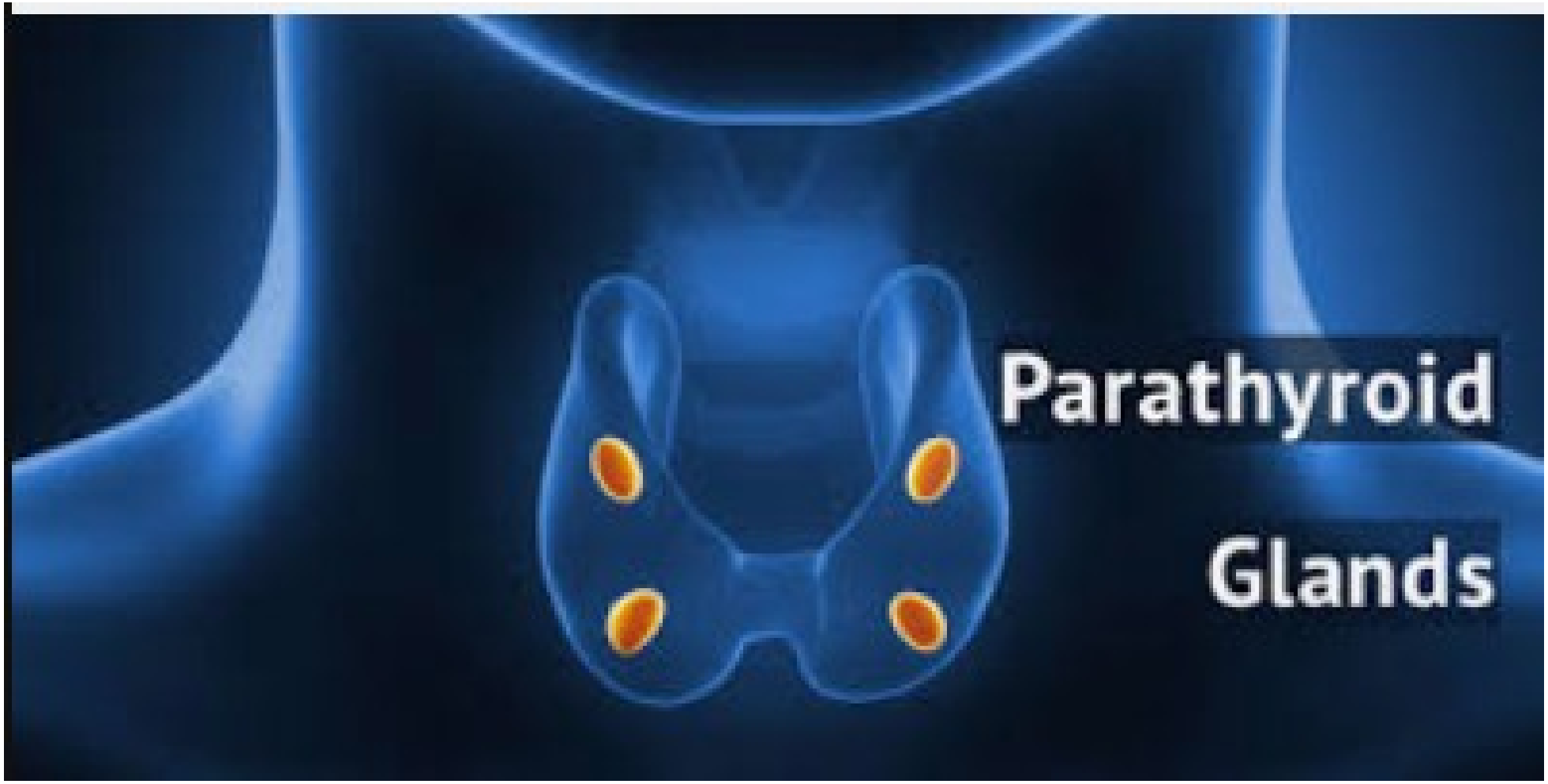
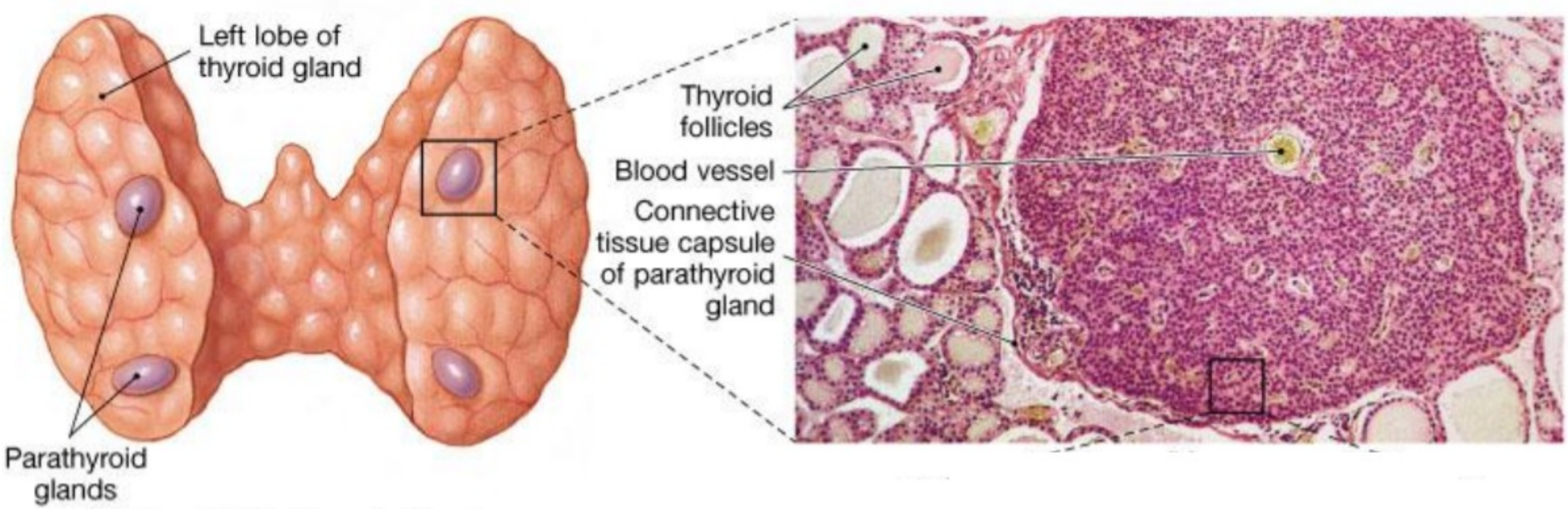
The Parathyroids, the Thyroid, and Ca^{2+} Regulation



Parathyroid Location and Structure



- The Parathyroids are separate endocrine glands embedded within the thyroid tissue.
- Produce *Parathyroid Hormone (PTH)*

Learning Aim: Describe the structure and location of the parathyroid glands.

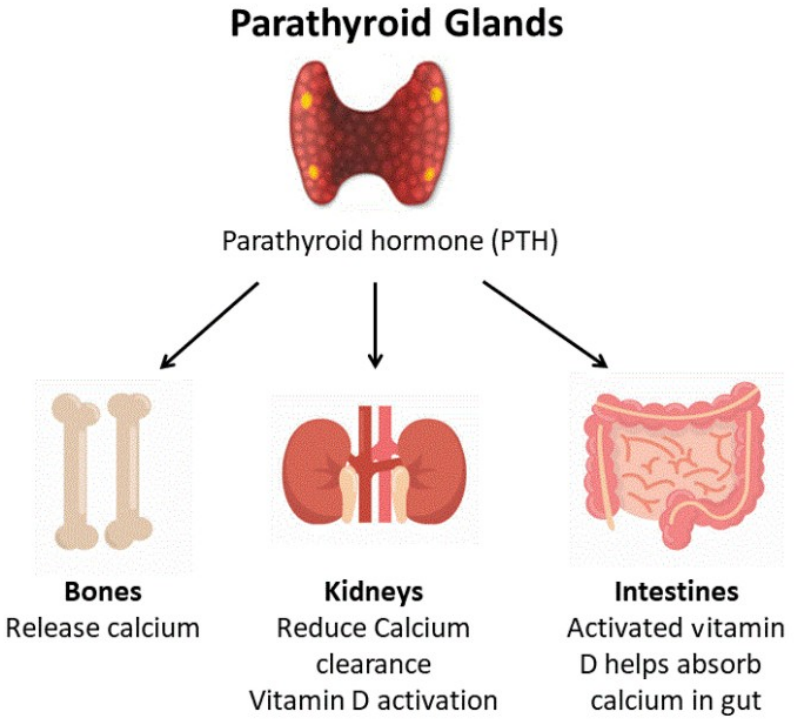
Parathyroid Hormone (PTH)

- Produced when Ca^{2+} levels in the blood become too low.

- Increases Ca^{2+} concentration in the blood by:
  - ✓ Increasing calcium reabsorbed into blood from renal tubules
  - ✓ Increases the amount of Ca^{2+} absorbed by the intestines from food
  - ✓ Triggers Ca^{2+} to be released from the bone matrix into the blood.

- Works in balance with Thyrocalcitonin to maintain Calcium ion homeostasis

- Also has a role in activating Vitamin D in the kidneys to increase Calcium absorption from food.



Learning Aim: Explain the effects parathyroid hormone (PTH) has on the body

The Thyroid and Thyrocalcitonin

- The thyroid gland also makes *Thyrocalcitonin* which is produced if Ca^{2+} levels in the blood are too high. **think back to Yr 11. Why would high blood Ca^{2+} be a problem?*

- Thyrocalcitonin reduces Ca^{2+} in the blood by:

  - ✓ Preventing Ca^{2+} from being reabsorbed by the kidney (so more is excreted in urine).
  - ✓ Promoting Ca^{2+} deposition in the bones (laying calcium down into the bone matrix).
  - ✓ Decreasing Ca^{2+} absorption by the intestines.



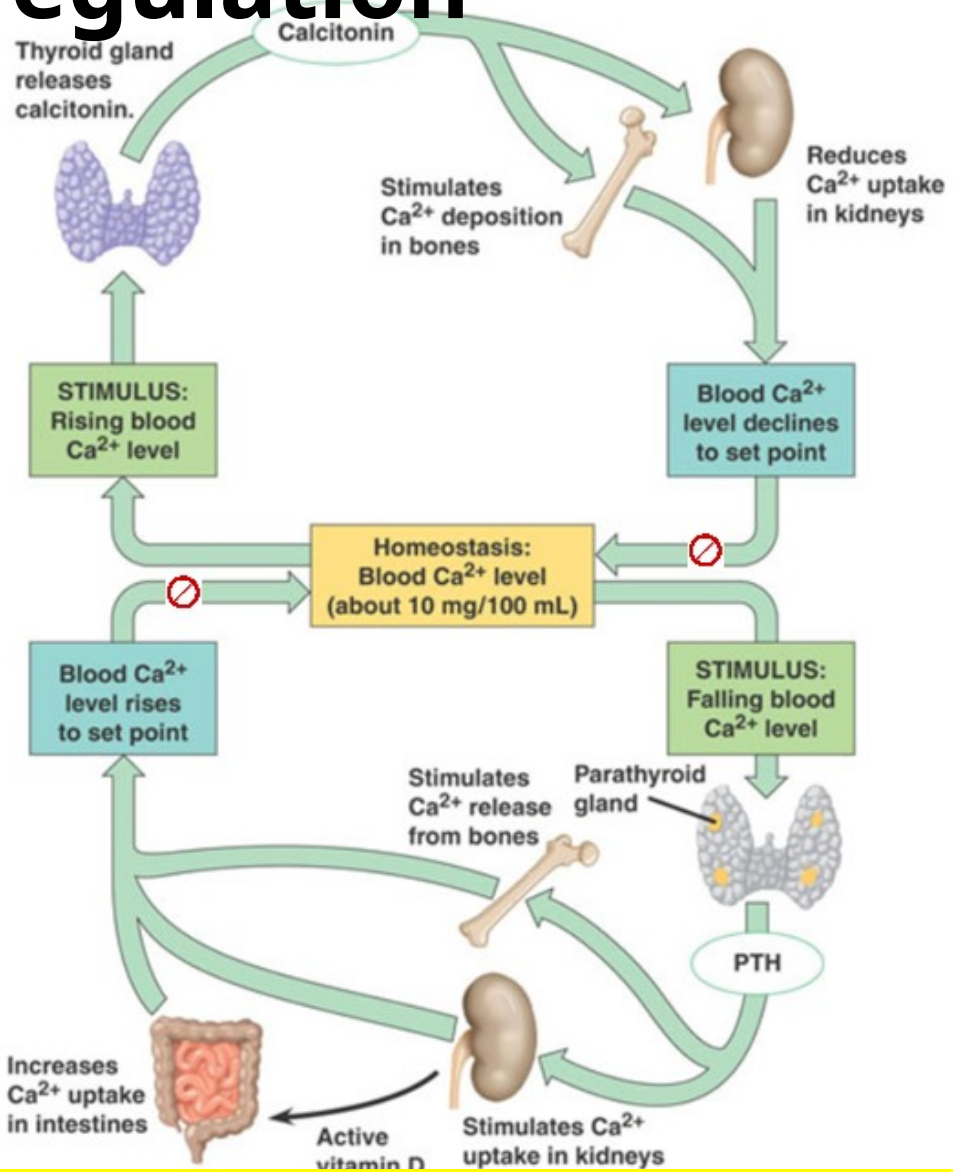




**note: Thyrocalcitonin works in balance with Parathyroid Hormone (PTH) produced by the Parathyroid.*

Learning Aim: Explain the effects thyrocalcitonin has on the body

The Thyroid, the Parathyroids and Blood Ca^{2+} regulation



When blood calcium levels rise:

1: The thyroid gland releases thyrocalcitonin

2: Thyrocalcitonin:

- Stimulates Ca^{2+} deposition in bones
- Reduces Ca^{2+} uptake in kidneys
- Inhibits Ca^{2+} absorption by intestines

3: When calcium levels stabilise to the set point, negative feedback prevents further release of thyrocalcitonin.

When blood calcium levels fall:

1: The parathyroid glands release PTH

2: PTH (Parathyroid hormone):

- Stimulates Ca^{2+} release from bones
- Increases Ca^{2+} uptake (resorption) in kidneys
- Increases Ca^{2+} uptake by intestines.

3: When calcium levels stabilise to the set point, negative feedback prevents further release of PTH.

Learning Aim: Explain how the body maintains constant levels of calcium ion in the blood using thyrocalcitonin and PTH.